

6. Electricity in the UK is generated in a variety of ways. Most of our electricity is produced by burning fossil fuels, mainly gas and coal. When deciding which type of power station to build, it is important to consider the environmental problems they cause.

(a) Tick (✓) the **two** correct statements below.

[2]

Burning fossil fuels adds to climate change

☐

Nuclear power stations emit lots of carbon dioxide when used

☐

Tidal barrages damage marine habitats

☐

Waste from gas power stations is radioactive

☐

Wind power causes acid rain

☐

(b) The table below shows the gases released when the same mass of different fossil fuels are burned.

| Fossil fuel | Emissions of polluting gas (units) |                 |                |
|-------------|------------------------------------|-----------------|----------------|
|             | Carbon dioxide                     | Carbon monoxide | Sulfur dioxide |
| coal        | 208 000                            | 208             | 2 591          |
| oil         | 164 000                            | 33              | 208            |
| gas         | 11 700                             | 40              | 1              |

Explain why coal has the greatest effect on global warming.

[2]

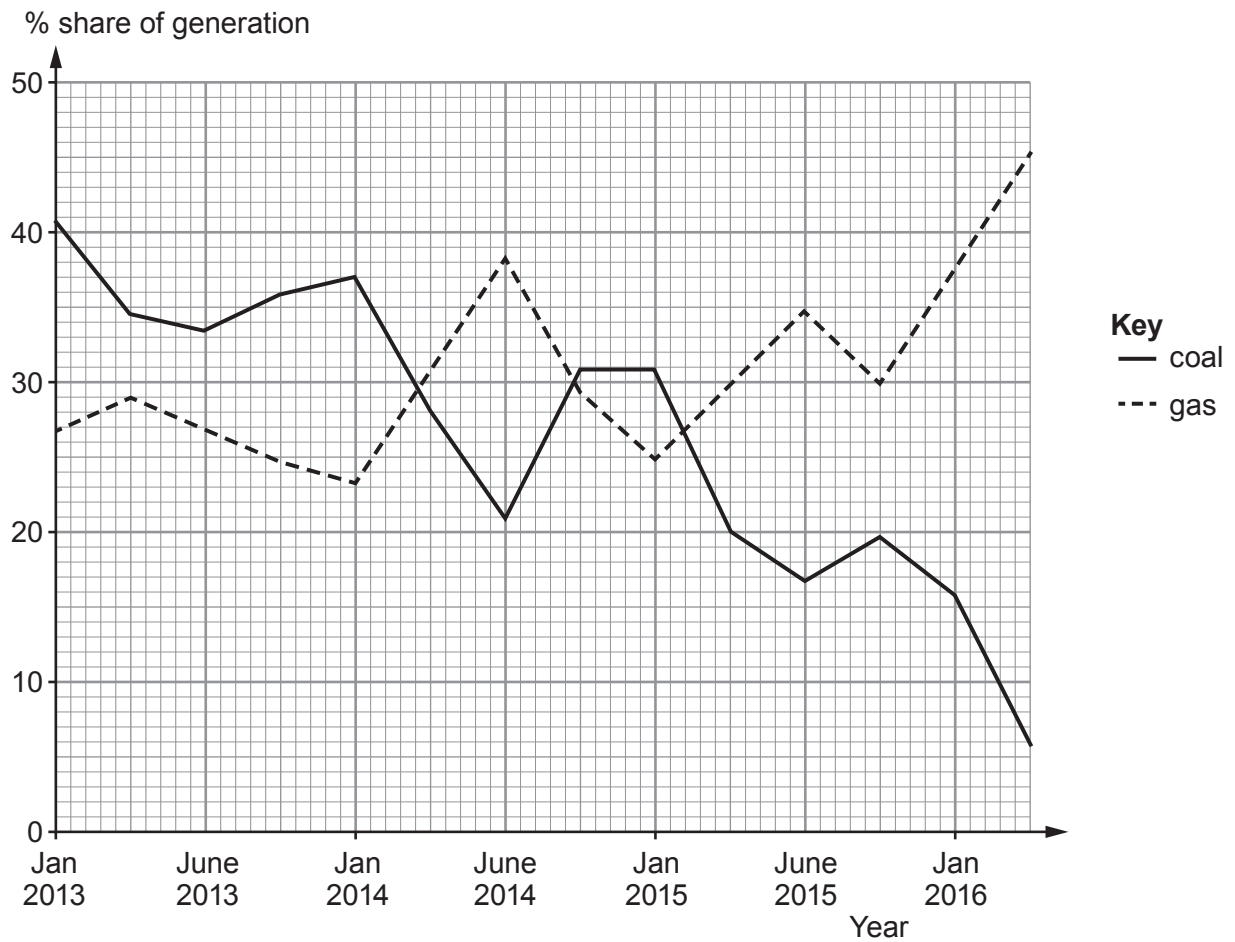
.....

.....

.....



- (c) Coal and gas are both used in power stations to generate electricity. The diagram below shows how the percentage share of electricity generation from coal and gas has changed since 2013.



Use the graph to compare how the percentages of electricity generated from coal and gas changed during the time shown. [2]

.....

.....

.....



- (d) The UK is trying to increase the percentage of electricity generated by renewable sources such as wind.

Between April 2015 and April 2016 many new wind farms and other renewable power stations were built.

The table below shows the percentage of electricity generated by renewable sources in April 2015 and in April 2016.

| Date       | % of electricity produced by renewable sources | % of electricity produced by non-renewable sources |
|------------|------------------------------------------------|----------------------------------------------------|
| April 2015 | 25.4                                           | 74.6                                               |
| April 2016 | 24.9                                           | 75.1                                               |

- (i) Use the data in the table to compare the percentage of electricity produced by renewable sources in April 2015 and April 2016. [1]

.....

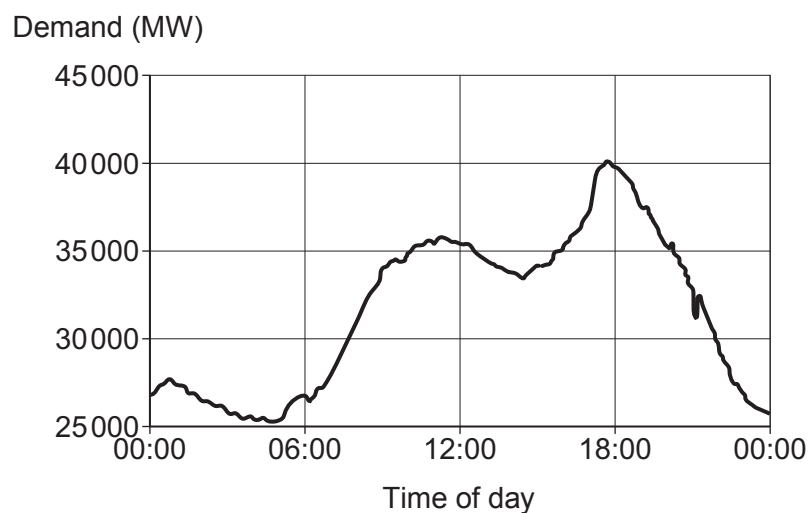
- (ii) Is your answer to (d)(i) what you expected? Give a reason for your answer. [1]

.....

.....



- (e) As well as generating electricity the UK also imports electricity from France. The diagram shows how demand for electricity varies throughout one day for the UK.



- (i) Give a reason why electricity is imported into the UK. [1]

.....

- (ii) At what time of day is electricity most likely to be imported into the UK from France? [1]

.....

